

Sravan Balaji

Email: balajsra@umich.edu Phone: (248) 417 - 0955 LinkedIn: sravan-balaji Website: sravanbalaji.com

SUMMARY

U.S. Citizen with active security clearance. Experienced roboticist with a strong multi-disciplinary background interested in working on autonomous systems in space, air, ground, and maritime domains.

WORK EXPERIENCE

- Metron**
Software Engineer II - Reston, VA
Software Engineer I - Reston, VA
 - Designed autonomous UUV control algorithms, implemented software in C++, tested in simulation, and integrated software onto hardware platforms for in-water testing

Apr. 2023 - Present
Nov. 2024 – Present
Apr. 2023 – Nov. 2024
- MITRE**
Intermediate Autonomous Systems Engineer - McLean, VA
 - Extended existing simulator functionality to allow for training of multi-agent reinforcement learning systems
 - Implemented terrain extraction from a bounding volume hierarchy data structure for UAV path planning
 - Setup, debugged, and demonstrated sensor reading and capabilities of the Hello Robot Stretch Mobile Manipulator for a healthcare application

May 2021 - Mar. 2023
Sep. 2022 – Mar. 2023
- Associate Autonomous Systems Engineer - McLean, VA
 - Led effort to take global path planning algorithm from research paper pseudo code, to Python prototype, to C++ and ROS implementation on a Clearpath Husky UGV; conducted testing in simulation with Gazebo and on the live platform; successfully avoided obstacles while applying desired safety factors
 - Developed a Python interface to translate existing drone messaging protocols to AirSim API calls for seamless transition to new simulation setup
 - Aided research effort into improving Test & Evaluation of Artificial Intelligence across the DoD

Jan. 2022 – Sep. 2022
- Graduate Navigation Intern - Remote
 - Designed and developed approach to identify implicit mitigation links between nodes in a directed acyclic graph for PNT Defense & Threat Library written in Python
 - Revamped and formalized software development process to improve visibility and organization of PNT Assurance project written in Julia
 - Placed 3rd in team-based machine learning intern hackathon; identified malicious URLs based on URL strings and information on age of domain; achieved 0.94 F1 score
 - Placed top 3 in intern AWS DeepRacer challenge; trained reinforcement learning model to control simulated autonomous vehicle around unseen race tracks

May 2021 – Aug. 2021
- University of Michigan - Ann Arbor**
NA 568 Grader - Remote
 - Graded assignments for NA 568: Mobile Robotics at the University of Michigan - Ann Arbor

Jan. 2021 - Apr. 2021
Jan. 2021 – Apr. 2021
- Rivian**
ADAS Controls Intern - Remote
 - Supported creation of Model-in-Loop framework for production controls software components; tested ~50 requirements, identified and resolved 7 issues
 - Integrated speed limit algorithm into production controls software for use with Simulink Coder C/C++ code generation; considerations made for embedded system processing limits and time delays
 - Modified MATLAB data processing scripts to work with new test logs; compared performance against requirements and simulation to determine controls parameters to tune

Jun. 2020 - Aug. 2020
Jun. 2020 – Aug. 2020
- Hughes Network Systems**
Software Development Intern - San Diego, CA

May 2019 - Aug. 2019
May 2019 – Aug. 2019

- Created Windows GUI app in C# employing .NET framework; automated mobile terminal configuration process to reduce user involvement, prevent errors, and decrease configuration time
- Implemented location based services on terminal software written in C; leveraged MQTT-SN protocol to provide low-cost method of sending GPS data to server

• University of Michigan - Ann Arbor

Sep. 2016 - Apr. 2019

Research Assistant in the Compliant Systems Design Lab - Ann Arbor, MI

Jan. 2019 – Apr. 2019

- Formulated and conducted an experiment to investigate applicability of digital image correlation (DIC) for analyzing strain in fiber reinforced elastomeric enclosures (FREEs)
- Manufactured a test setup to securely hold FREEs and control flow of compressed gas
- Created FREEs with varying parameters by hand to investigate their behaviour under pneumatic loads
- Developed a C++ program to analyze creep behavior of a FREE and generate plots

Intramural Sports Supervisor - Ann Arbor, MI

Sep. 2017 – Dec. 2018

- Voted Supervisor of the Year by peers for 2017-2018 school year
- Developed an excel macro to automate scheduling supervisor shifts; eliminated issues with transportation of people and equipment present in manual scheduling process
- Supervised intramural sports games at various facilities on campus; evaluated performance of officials to improve customer experience; resolved any disputes that arose

Intramural Sports Official - Ann Arbor, MI

Sep. 2016 – Dec. 2018

- Enforced the rules of the game for intramural soccer, flag football, and basketball matches between college students
- Managed flow of the game to ensure customer enjoyment and safety

• Rivian

May 2017 - Dec. 2018

Business Technology Intern - Plymouth, MI

May 2018 – Dec. 2018

- Championed effort to remove BOM data inconsistencies; developed Excel macro to summarize mass and cost data, alert PMs of incorrect data; resulted in faster detection and resolution of issues
- Introduced and designed an internal website to collect and display data from business systems utilizing Python and Django; worked with REST APIs to present reports of issues by severity
- Led effort to migrate IT team to a new service desk; worked with Jira Service Desk to automate triage and assignment of IT tickets; increased ticket resolution rate within SLA targets from 70% to 95%

Vehicle Integration Intern - Livonia, MI

May 2017 – Aug. 2017

- Created system architecture models to distribute and track requirements from system to component level
- Managed requirements in DOORS Next Generation and created custom reports to distribute requirements to sub-teams
- Developed a program in Java to summarize results of simulation; allowed users to modify inputs to see projected results of simulation to avoid additional testing and reduce costs

EDUCATION

• University of Michigan - Ann Arbor

Sep. 2016 - Dec. 2021

M.S. in Robotics; Focus in Acting; GPA: 3.982 - Ann Arbor, MI

Jan. 2021 – Dec. 2021

- Mobile Robotics; Design of Digital Control Systems; Self Driving Cars; Robotic Systems Lab

B.S.E. in Mechanical Engineering; Focus in Controls; GPA: 3.776 - Ann Arbor, MI

Sep. 2016 – Dec. 2020

- Automatic Control; Modeling, Analysis, and Control of Dynamic Systems

B.S.E. in Computer Science; Focus in Intelligent Systems; GPA: 3.776 - Ann Arbor, MI

Sep. 2016 – Dec. 2020

- Autonomous Robotics Lab; Machine Learning; Computer Vision

SKILLS

- **Programming:** C/C++, Python, MATLAB, Julia, JavaScript, C#, Java
- **Development:** GNU/Linux, Git, Docker, Visual Studio Code, GNU Emacs, Vim, JetBrains IDEs
- **Libraries/Frameworks:** ROS, Protocol Buffers, move_base, NumPy, OpenCV, PyTorch
- **Robotics/Autonomy:** Controls, Path Planning, Kalman Filters, SLAM, GPS, Forward & Inverse Kinematics, Machine Learning, Computer Vision
- **Modeling and Manufacturing:** 3D Printing, SolidWorks, CATIA
- **Languages:** English (first language), French (conversational), Tamil (conversational)